EST PROJECT 21 · MID-EVALUATION

Cascading Climate Stress

Does one climate extreme leave nature weaker against the next?

Case 1: SE Australia (drought first) · Case 2: Pacific Northwest (heat first)

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PROBLEM STATEMENT

Climate extremes now often come one after another. Does the first one leave ecosystems hurt more by the next, and slower to recover?

Most studies look at one event at a time. We built one method from free satellite and weather data, and tested it on real chains of events that start with different extremes.

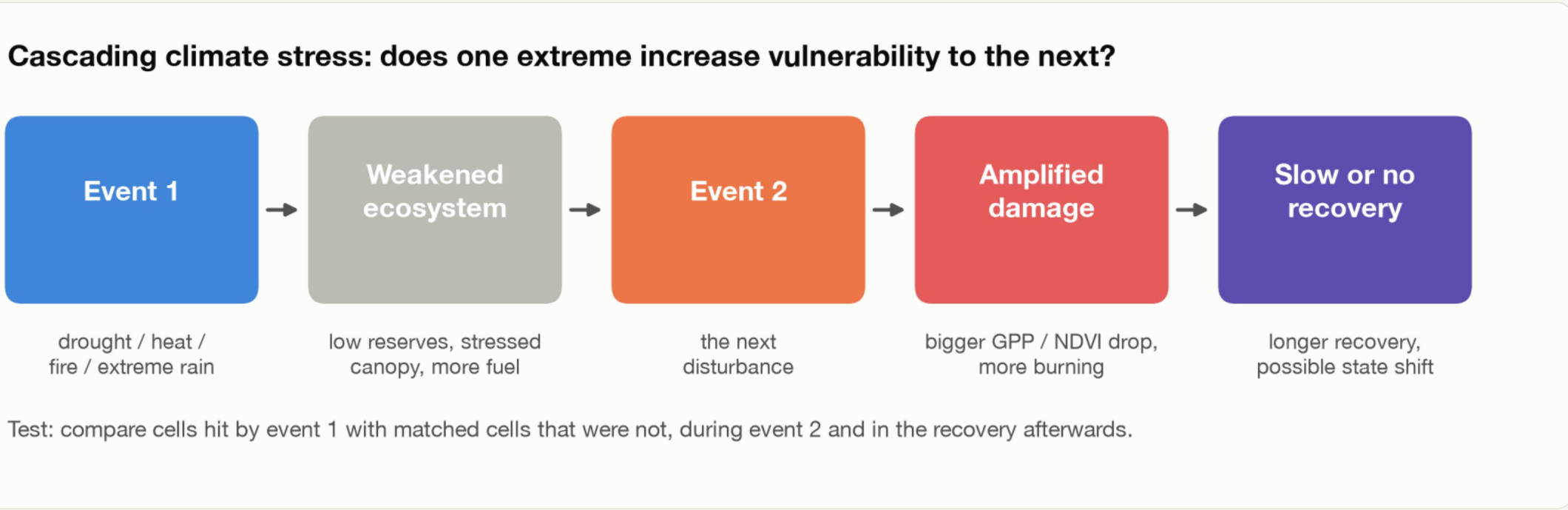
PROBLEM STATEMENT

Five hypotheses we test

#	Hypothesis	Tested by
H1	Grid cells (≈10 km squares) hit by event 1 are hurt more by event 2 than similar cells that were not hit	Module C: fair comparison, fire model
H2	The worse event 1 was, the bigger the damage, maybe with a tipping point	Module E: severity vs effect
H3	Cells hit by both events recover more slowly, or not at all	Module D: recovery
H4	The effect of event 1 lasts for months or years	Module D: delayed effects
H5	Some ecosystem types are hit harder than others	C and D, split by ecosystem

PROBLEM STATEMENT

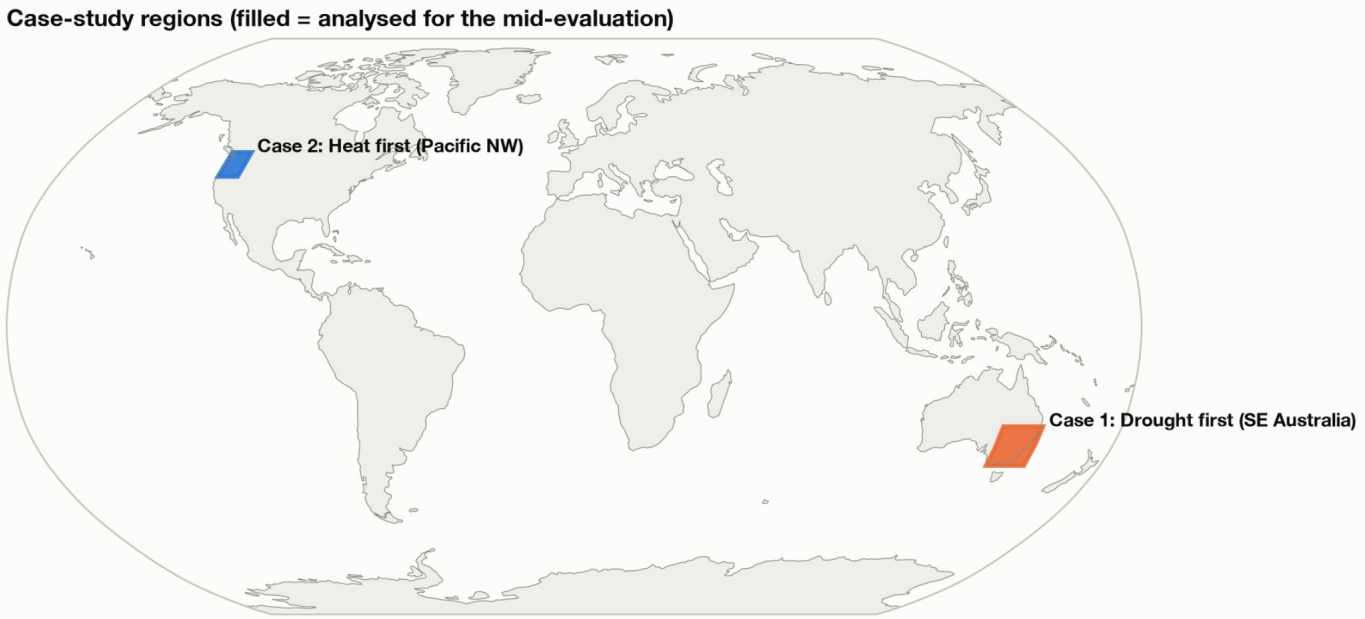
The cascade we are testing



We compare areas hit by event 1 with similar areas that were not, during event 2 and while they recover.

DATA USED

Two real cascades, each starting with a different extreme



Case 1 SE Australia: 2017–19 drought → 2019–20 heatwaves and Black Summer fires. **Case 2** Pacific NW: June 2021 heat dome → late-summer drought and fires.

DATA USED

All six datasets on one shared grid (≈10 km, monthly, 2001–2024)

Dataset	What it measures	What we use it for
ERA5-Land	Daily max temperature, air dryness, soil moisture, rain	Heatwaves, dryness, checking rainfall
SPEI	Drought index (3- and 12-month)	Where the drought was and how bad
CHIRPS	Rainfall	Rain vs normal; a check on ERA5
MODIS fire	Burned area, fire intensity	Where it burned and how hot
VIIRS (NASA FIRMS)	Fire detections (375 m)	A second, independent fire check
MODIS GPP / NDVI	Plant growth (GPP), greenness (NDVI)	How vegetation reacted and recovered
Extra: MODIS land cover, SRTM	Land cover, elevation	Removing farms and towns, pairing cells, ecosystem split

DATA USED

The data match what we know happened

−1.3

Average drought index over SE Australia in 2018 and 2019: the record drought

5.6 Mha

Burned in the Case 1 region, Jul 2019 to Mar 2020 (MODIS)

r = 0.88

Two rain datasets (CHIRPS, ERA5) agree, Case 1 (0.93 in Case 2)

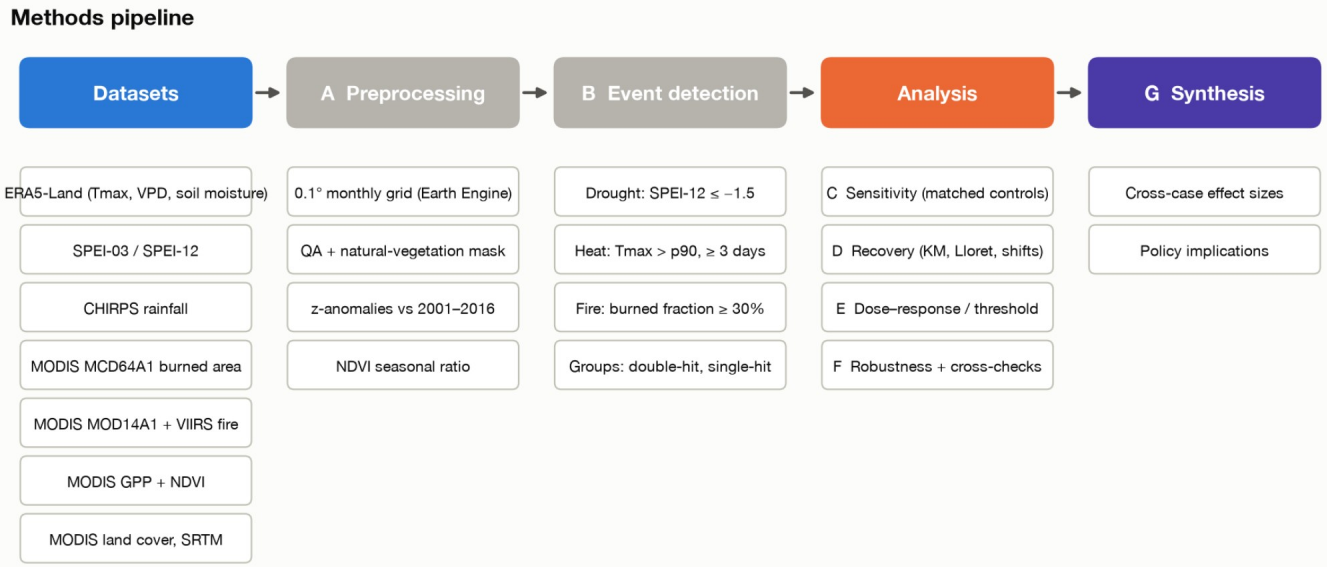
ρ = 0.76

Two fire satellites (MODIS, VIIRS) agree, Case 1 (0.63 in Case 2)

The wet La Niña years (2010, 2022) and the 2019 heat record also show up. Each dataset agrees with an independent one, so we can trust them together.

METHODOLOGY

One pipeline for every case: Modules A to G



To add a new case we only change the region and the event dates in one settings file. All the code is shared.

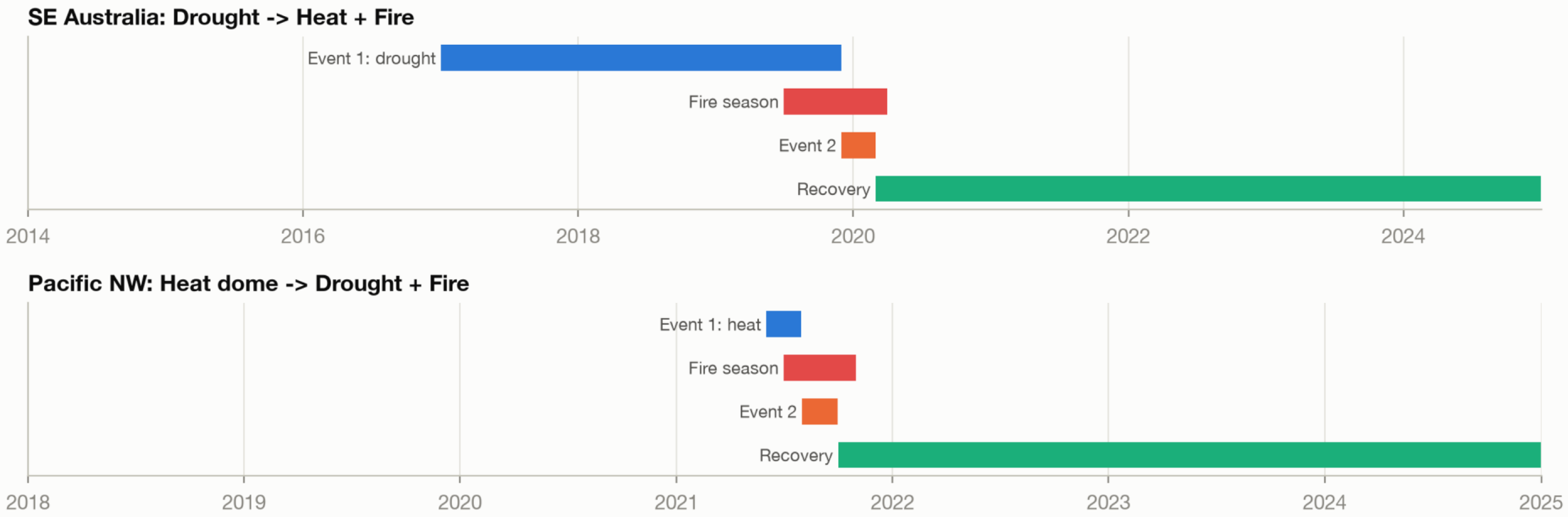
METHODOLOGY

How we define events and groups

Item	Definition
Drought (Case 1)	Drought index (SPEI-12) reached −1.5 or lower during 2017–19
Heat dome (Case 2)	Hottest day, 25 Jun to 2 Jul 2021, at least 2 standard deviations (SD) above normal. Heatwave = 3+ days hotter than the usual top 10%
Burned	At least 30% of the cell burned in the fire season (60% for recovery). Cells that burned in the 3 years before are left out
Response	How far plant growth (GPP) and greenness (NDVI) fell below normal (2001–2016) in event 2, in unburned cells
Fair comparison	Each hit cell is paired with a nearby unhit cell with the same land cover and similar elevation and event-2 stress
Recovered	Greenness (or growth) back to 90% of normal for 3 months in a row
Statistics	Standard tests with 95% confidence ranges; neighbouring cells grouped so we don't overstate certainty

METHODOLOGY

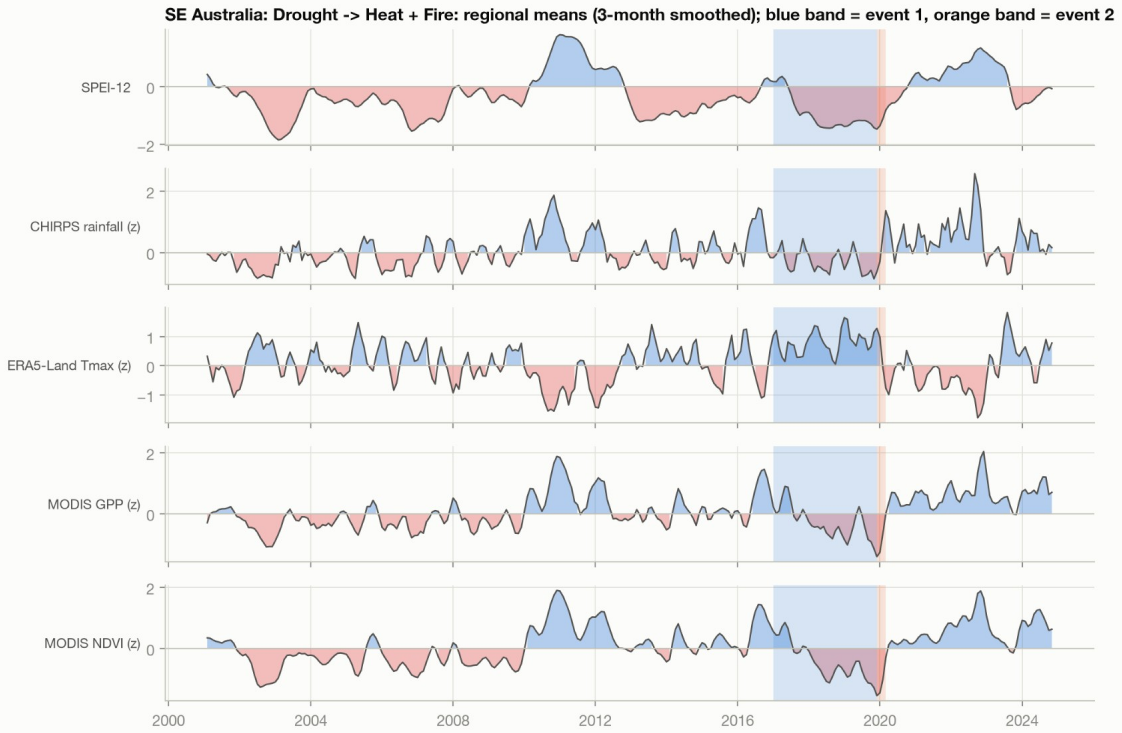
When each event happened



Case 1 is slow: years of drought, then fire. Case 2 is fast: a one-week heat dome, then drought a few weeks later.

RESULTS · CASE 1

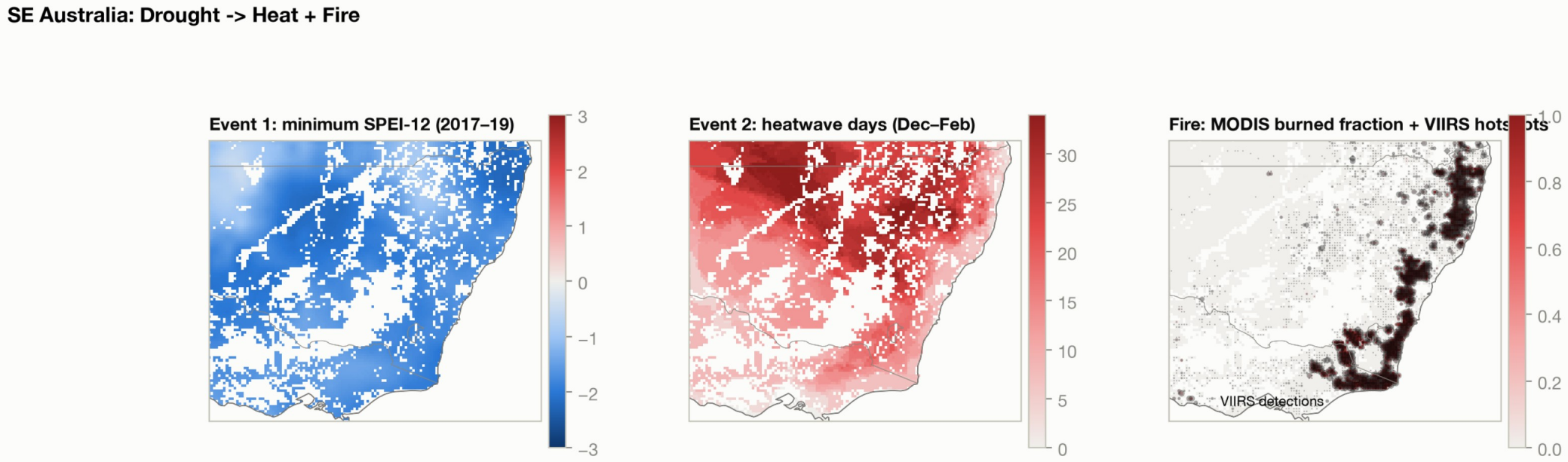
SE Australia: drought, then heat and fire



- Drought index and rain fall for three years (blue band)
- Temperature peaks in summer 2019–20 (orange band)
- Plant growth and greenness hit record lows during the fires
- Strong bounce-back in the wet La Niña years 2020–22

RESULTS · CASE 1

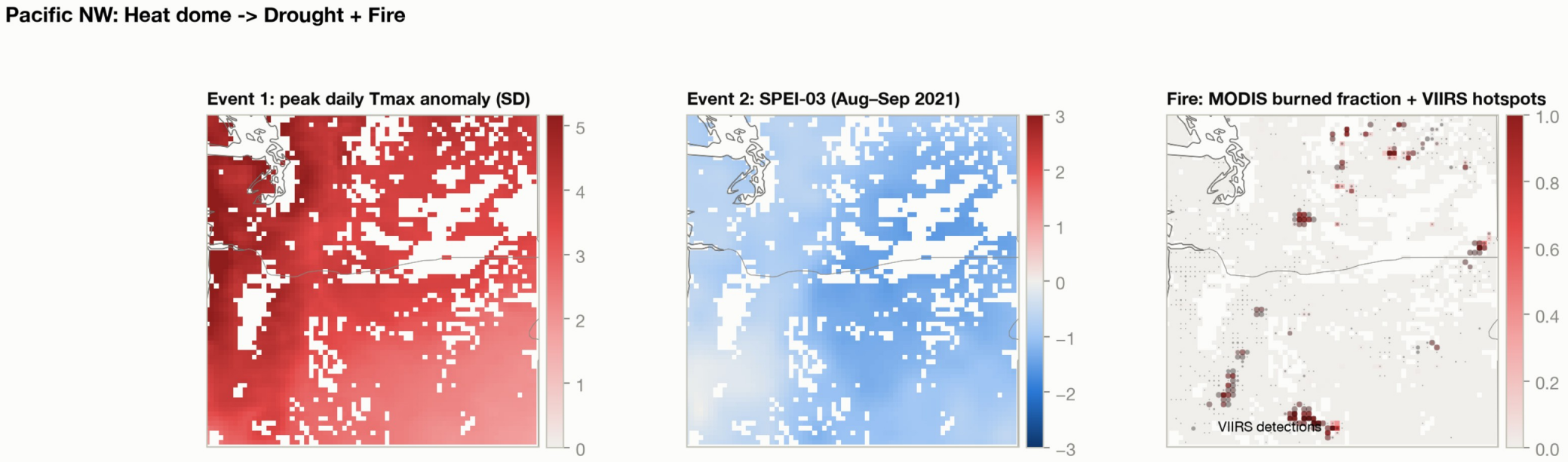
Where the drought, heat and fire hit



The drought covered almost the whole region. Heatwaves were worst inland. Fires were mostly in the coastal and mountain forests, and both fire satellites agree on where.

RESULTS · CASE 2

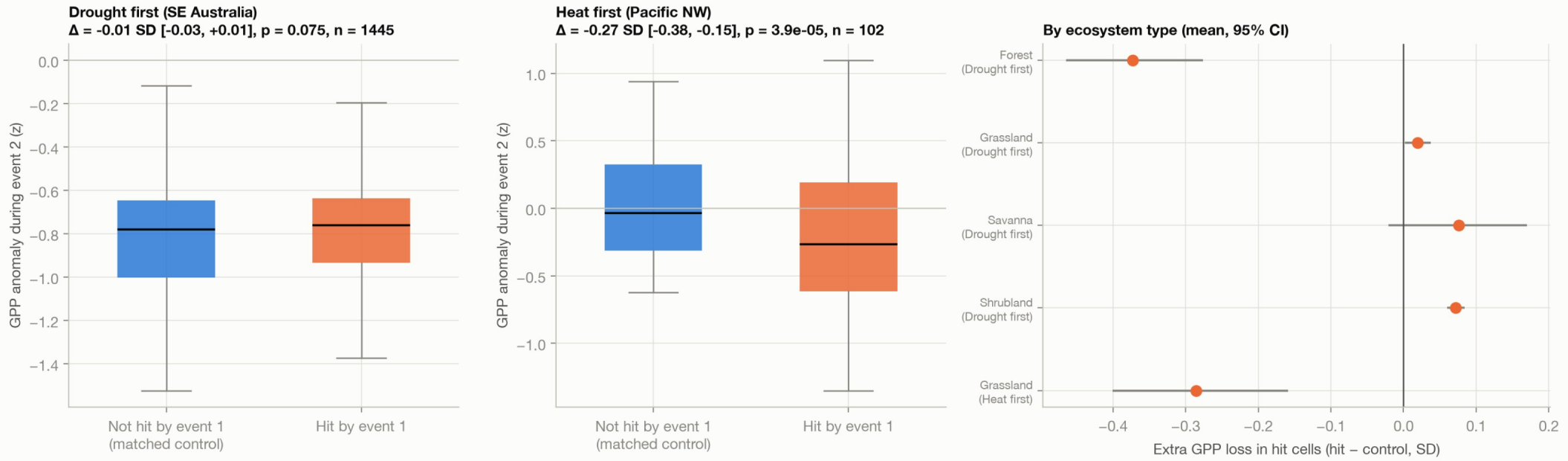
Pacific NW: the heat dome hit 97% of cells



The typical cell peaked 3.7 SD above normal. Drought followed almost everywhere in late summer. The 2021 fires were small and scattered: only 37 burned cells once we leave out areas that burned in 2020.

RESULTS · H1 SENSITIVITY

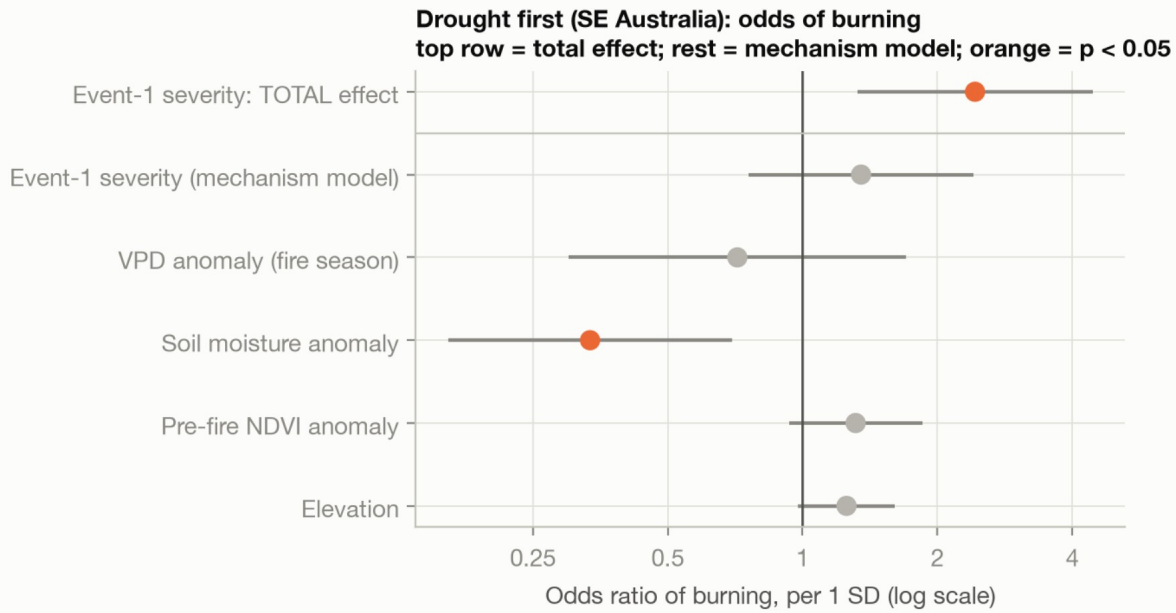
Did event 1 make plants lose more growth in event 2?



Case 1: no (−0.01 SD, p = 0.08), except in forests (−0.37 SD). **Case 2: yes, a little**: −0.27 SD [−0.38, −0.15], p < 0.001, but the result reverses with a stricter 3 SD cutoff.

RESULTS · H1 FIRE

Prior drought made fire much more likely

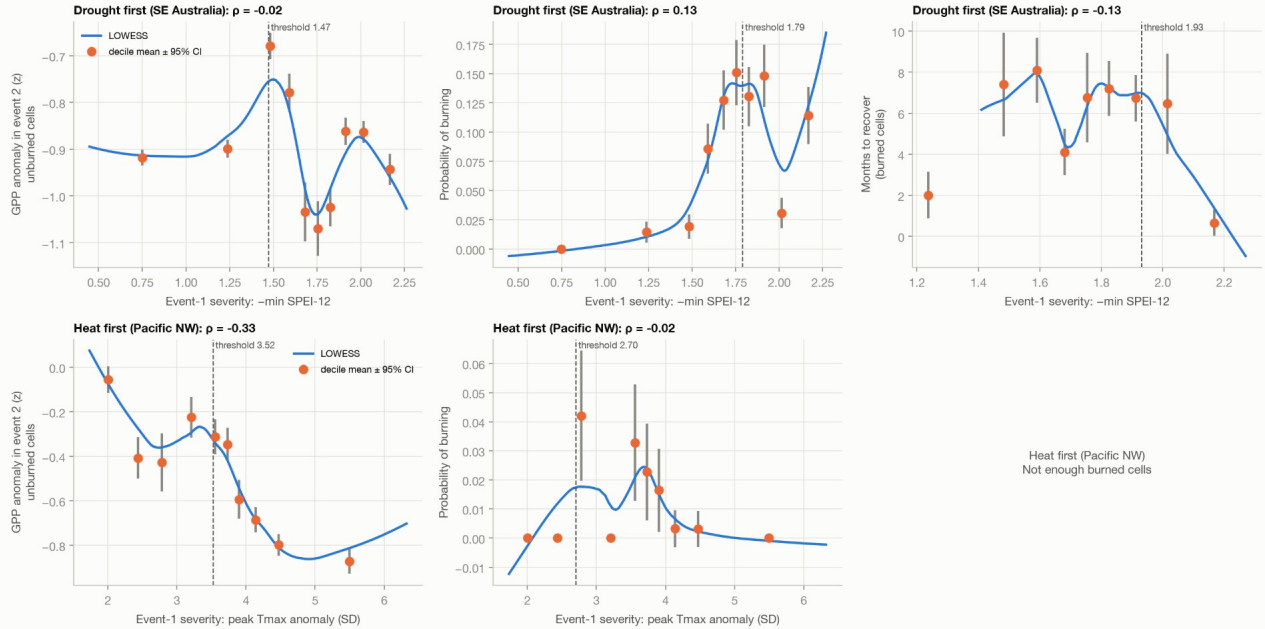


Heat first (Pacific NW)
Too few burned cells for a fire model

The odds of burning rose **2.4× for each 1 SD more drought** [1.3, 4.4], p = 0.004. In the driest third of forest cells 66% burned, vs 21% in the least dry third. Drought works by drying out the soil.

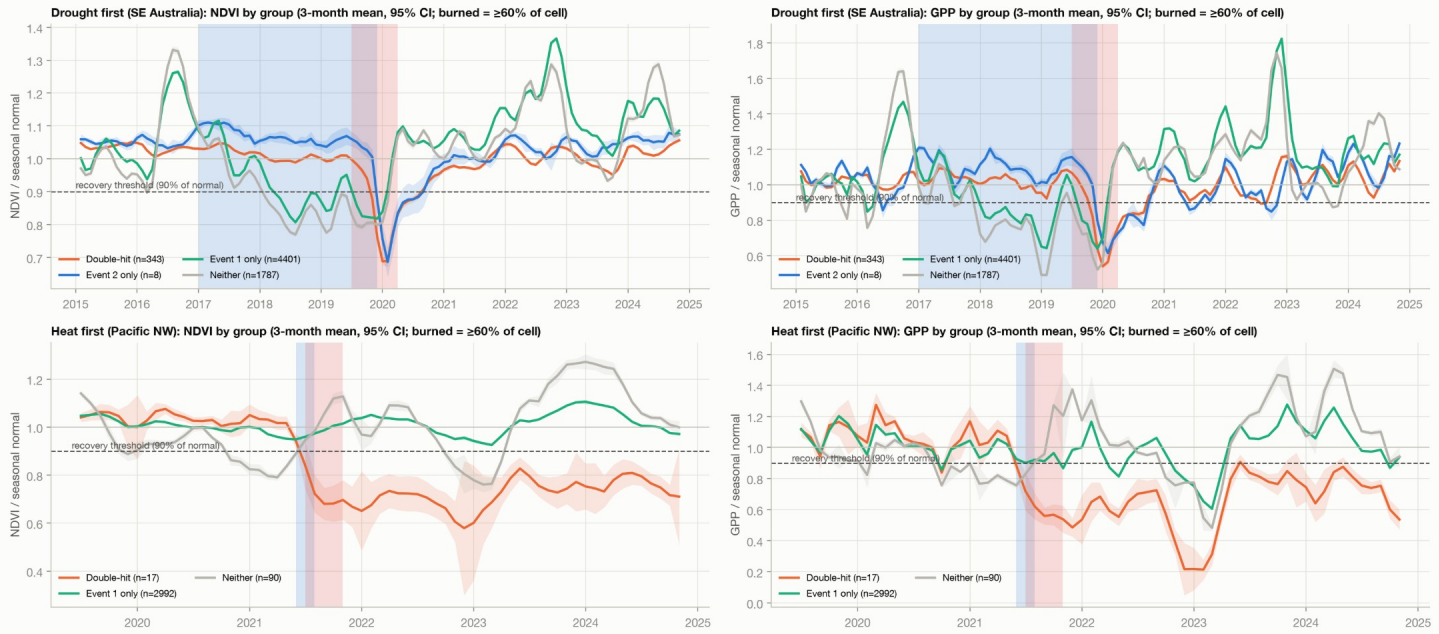
RESULTS · H2 SEVERITY VS EFFECT

Worse first events, bigger effects, up to a tipping point



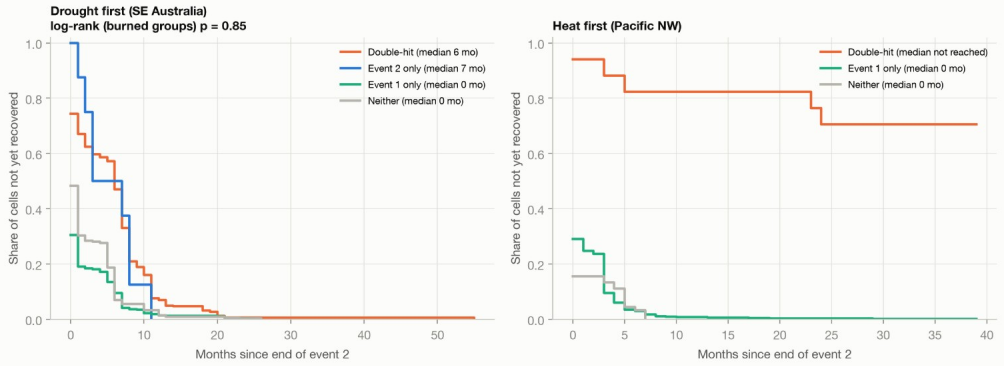
Case 1: the chance of burning rises as drought gets worse, up to SPEI ≈ −1.8, then flattens. Case 2: plants lose more growth the stronger the heat dome (ρ = −0.33), faster past ≈ 3.5 SD.

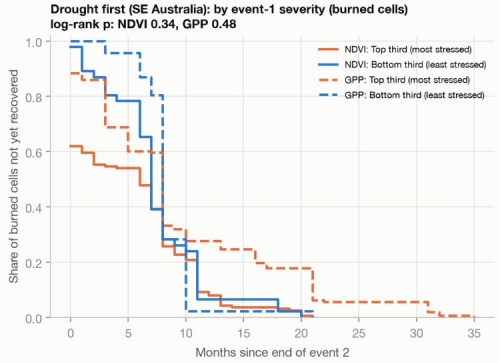
Recovery: fast in eucalyptus, missing in burned conifers



- Australia's burned forests: greenness back to 90% in ~6 months, growth in ~8
- Leaves come back before full growth does
- Pacific NW burned areas: greenness still ~30% below normal in 2024
- Unburned areas barely dipped

Time to recover did not depend on prior drought



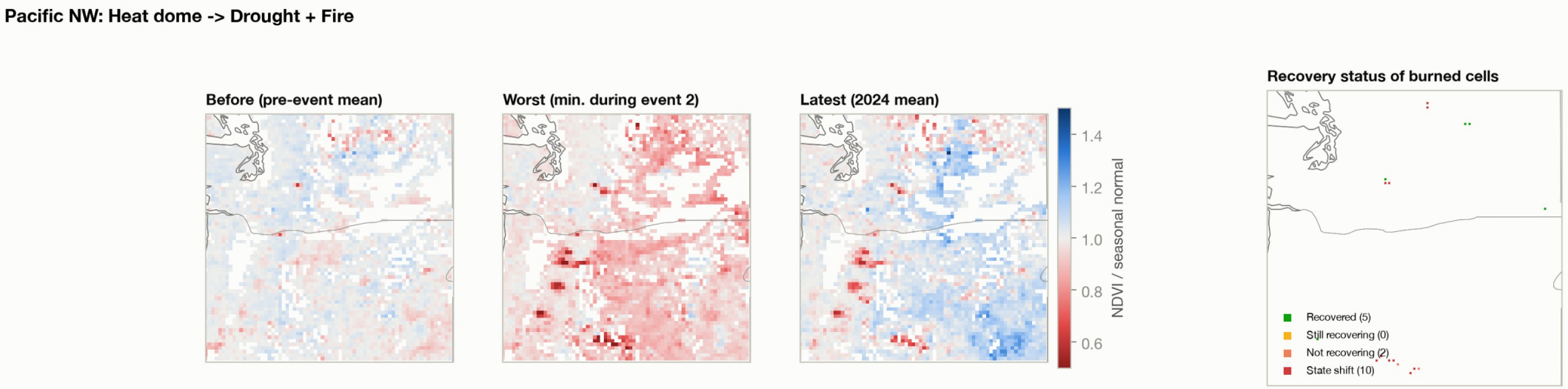


Heat first (Pacific NW)
Too few heavily burned cells
to split by event-1 severity

- Case 1: burned areas recovered at the same speed after severe or mild drought (p = 0.34 greenness, 0.48 growth)
- All Case 1 areas recovered by 2024
- Case 2: most burned areas have still not recovered

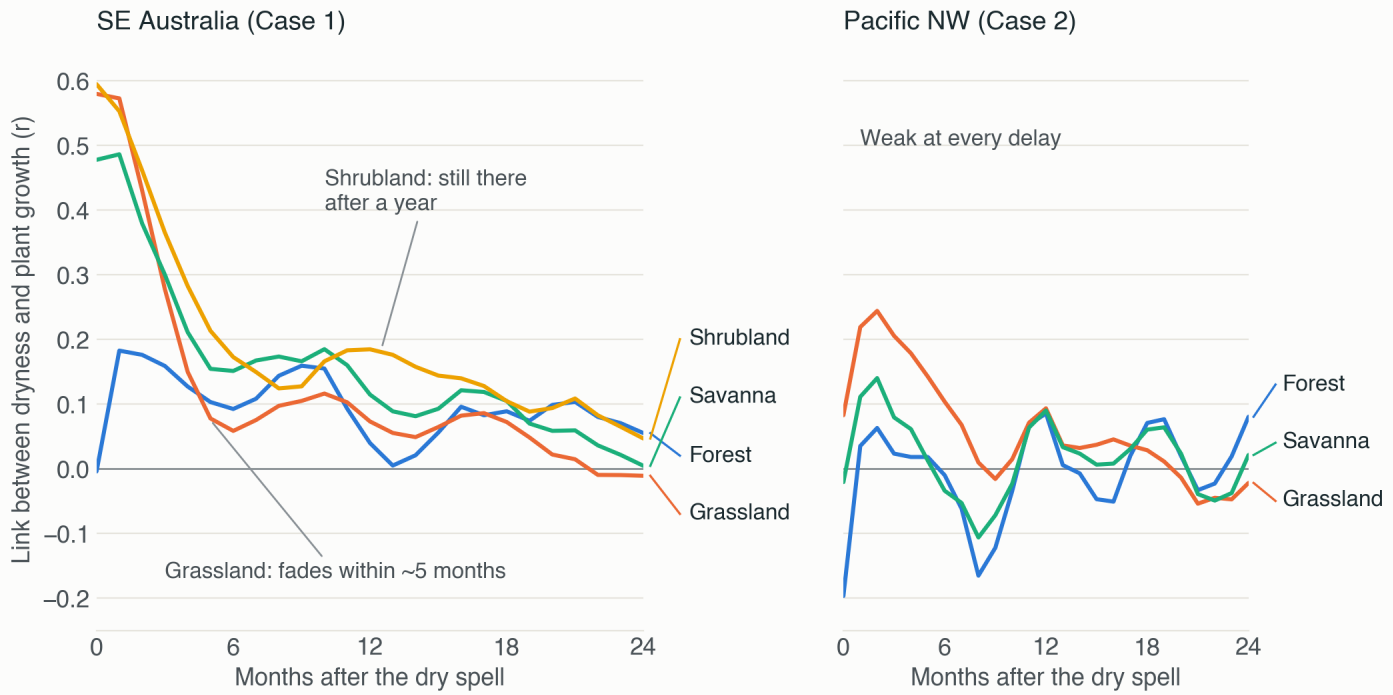
RESULTS · H3 RECOVERY

Pacific NW: signs of a lasting change



Of 17 heavily burned cells, 10 (59%) stayed below 90% of normal for 2+ years (a "state shift") and only 5 recovered. A small sample, but very different from Australia.

How long does a dry spell keep affecting plants?



- Australia: a dry spell cuts growth in grass, shrub and savanna right away ($r \approx 0.5$ to 0.6)
- Grassland bounces back within ~5 months; shrubland is still affected a year later
- Forests barely react to short dry spells
- Pacific NW: very little memory

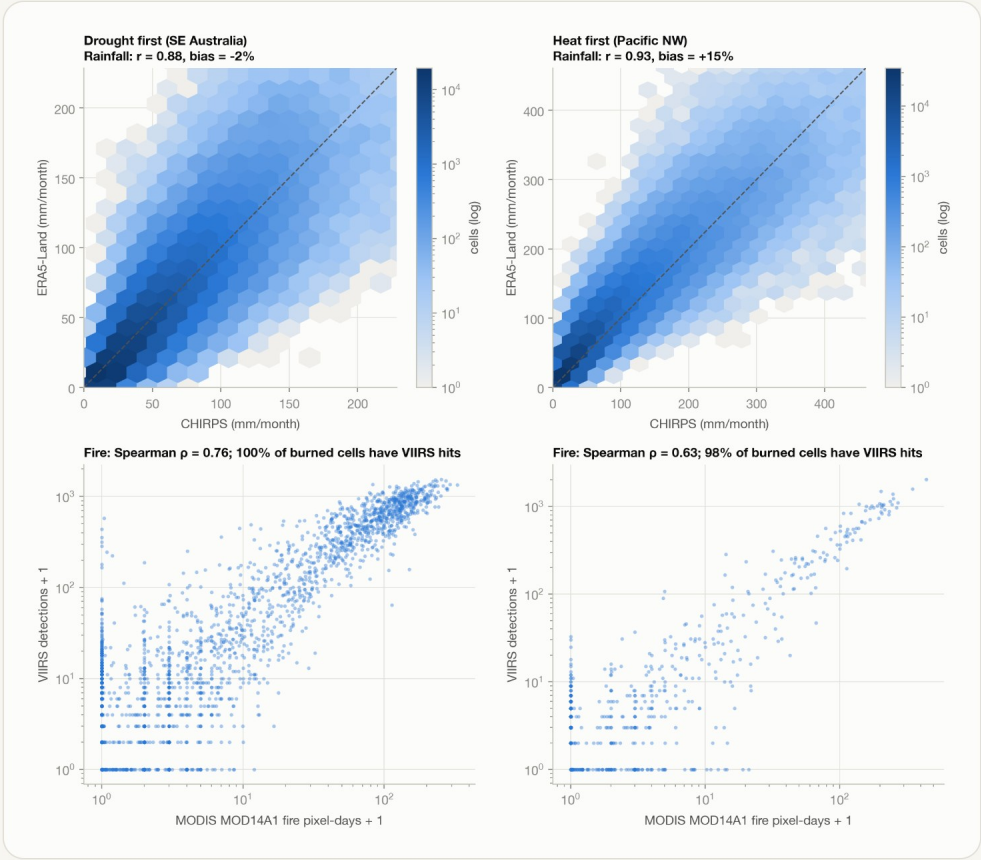
RESULTS · H5 ECOSYSTEM TYPE

In Australia, only forests show the cascade

Ecosystem	Extra growth loss after drought	Burned: driest vs least dry third	Months to recover after fire
Forest	−0.37 SD (significant)	66% vs 21%	~7
Savanna	None	4% vs 2%	~6
Grassland	None	1% vs 0%	Barely dipped
Shrubland	Slightly less loss (+0.07)	0% vs 0%	Too few burned

Pacific NW: grassland lost extra growth after the heat dome (−0.29 SD), and 12 of 14 burned forest and open-woodland cells had not recovered by 2024.

The datasets agree; some results depend on cutoffs

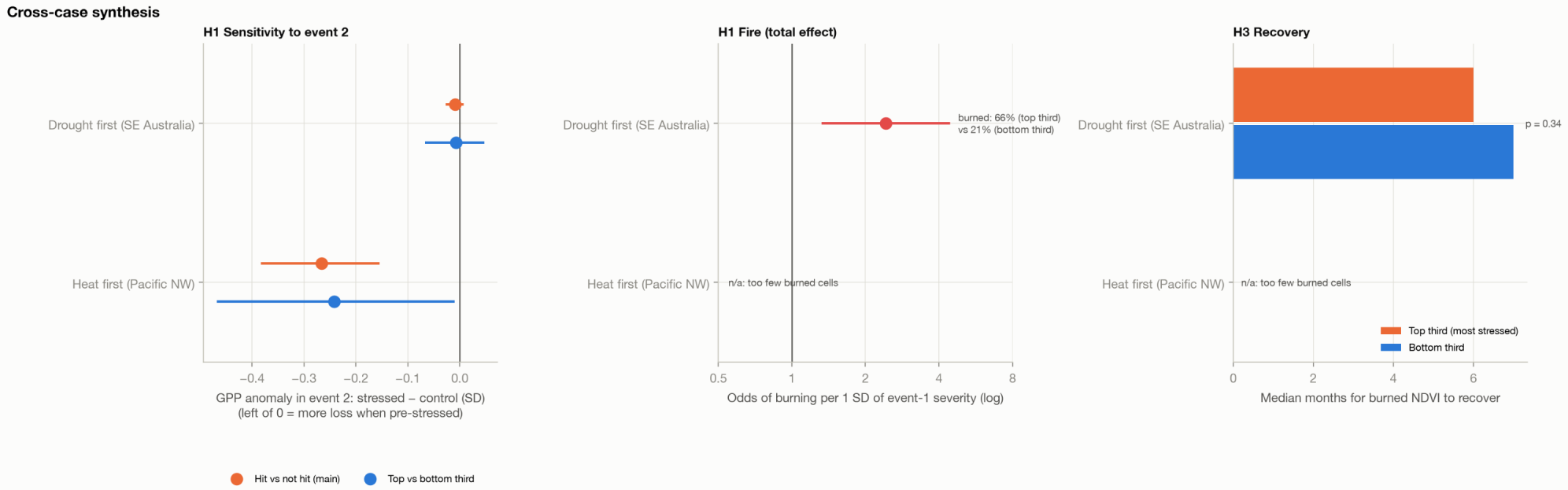


Check	Case 1	Case 2
Rain: CHIRPS vs ERA5 (r)	0.88	0.93
Rain difference (ERA5 – CHIRPS)	–2%	+15%
Fire: MODIS vs VIIRS (ρ)	0.76	0.63
Drought index vs soil moisture (r)	0.65	0.49
Extra growth loss, main cutoff	–0.01 (not sig.)	–0.27
Extra growth loss, other cutoff	+0.05	+0.14

The fire result holds both ways. The growth result reverses when we change the cutoff, so we only call it suggestive.

OBSERVATIONS

Both cases side by side



The chain shows up through **fire** after drought, and weakly through **plant growth** after heat. Recovery depends more on the type of ecosystem than on which extreme came first.

OBSERVATIONS

What the first two cases show

Hypothesis	Case 1: drought first	Case 2: heat first
H1 plant growth	Not supported: no extra loss in the heatwave (−0.01 SD)	Suggestive: −0.27 SD extra loss in the drought, but it changes with the cutoff
H1 fire	Supported: odds of burning ×2.4 per SD of drought (p = 0.004)	Can't test: only 37 burned cells
H2 tipping point	Burn risk rises until SPEI ≈ −1.8, then flattens	Growth loss speeds up past ≈ 3.5 SD of heat (uncertain)
H3 recovery	Fast and complete: greenness ~6 months, growth ~8; prior drought made no difference	Burned conifers (17 cells): 71% not recovered by 2024, 59% lasting change
H4 memory	Dry-spell effect lasts ~5 months in grass, about a year in shrub and savanna	Very little memory
H5 ecosystem	Forests: −0.37 SD extra loss and most burning; other types little or none	Grassland: −0.29 SD; burned conifers not recovering

OBSERVATIONS

What this means for policy (first draft)

Drought → fire

Use drought maps (SPEI, soil moisture) as an early fire warning. Clear fuel first where the drought was worst.

Heat → drought

After a heatwave, treat the hardest-hit areas as high fire risk for the rest of the season and watch them closely.

Recovery

Focus replanting on forests that don't regrow by themselves, like burned conifers, using maps of what has recovered.

Plan for chains of events. Looking at one extreme at a time underestimates the risk.

OBSERVATIONS

Limitations, and what we did about them

1. Almost everything was hit

74% and 97% of cells were hit by the first extreme, so few untouched areas were left to compare with.

What we did: paired each hit area with the most similar nearby area, and compared the worst-hit third with the least-hit third.

2. Few fires in the Pacific NW

Only 37 cells burned there, too few to test whether the heat dome raised fire risk.

What we did: we treat those results as early signs, not proof. Case 3 (California 2020) adds many more fires.

3. One result depends on the cutoff

The extra plant-growth loss reverses if we change how dry or hot counts as "hit".

What we did: we report both cutoffs and only call that result suggestive. The fire result holds either way.

4. Links, not proof of cause

Satellite records show what happened together. They can't fully prove one thing caused the other.

What we did: like-for-like comparisons, plus a physical explanation: drought dries the soil, and dry land burns.

NEXT STEPS

What's done, and what comes before the final evaluation

Done (mid-evaluation)

- Full pipeline with all datasets
- Cases 1 and 2 through Modules A–G
- Checks between datasets
- First summary and policy draft

Before the final

- Case 3: California fires 2020 → drought 2021
- Optional Case 4: heavy rain → fire
- More checks with other cutoffs
- Final summary, policy and report

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Thank you

Questions are welcome.

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